NR2DT-01: Step 1 — Discover

> **Series:** NR2DT — New Relic to Dynatrace Migration Steps | **Notebook:** 1 of 10 | **Created:** April 2026 | **Last Updated:** 11/03/2026

Overview

Goal of this step: produce a complete inventory of the source New Relic account, a translation-confidence report, and a defensible effort estimate. Nothing else moves until discovery is complete.

This is procedural. For the *why* behind each artifact and the deeper translation theory, see **NRLC-02** (NRQL→DQL Translation) and the relevant NRLC component notebook for each entity class.

Sprint 1.337 (April 2026) Updates Affecting NR2DT

Sprint 1.337 introduces three changes worth surfacing during the discovery step:

1. **OneAgent primary fields/tags at source** — when designing the post-migration tag taxonomy (see NR2DT-03 Design), prefer OneAgent primary tags over OpenPipeline parsing where possible. From OneAgent 1.333, hosts set primary Grail tags with `--set-host-tag` / `oneagentctl --set-host-tag`; the key carries the `primary_tags.` prefix (`primary_tags.team`, `primary_tags.cost_center`, `primary_tags.env`).
2. **OTel `service.name` for SDv1 service names** — SaaS 1.337 lets the OTel `service.name` attribute name SDv1 services; Smartscape shows the name in the form `service.name (detected name)` (NRLC-01 § Sprint 1.337 covers the deeper detail).
3. **Settings v2 + Platform tokens** for new automation (NR2DT-04 Translate, NR2DT-09 Cutover scripts).

Engine impact: the Dynatrace-NewRelic conversion engine and lab-setup scripts in `topics/nr2dt/lab-setup/` were validated end-to-end on a live Gen3 sprint tenant 2026-04-20 — re-verify before each new engagement that the conversion engine handles sprint-337's API removals (Entity API `attributes`, Event API `metadata`).

> **Sources:** Primary Grail tags on OneAgent (DT docs) — *"Primary Grail field and tag enrichment requires OneAgent version 1.333+."*, What's new in Dynatrace SaaS 1.337 (DT docs) — *"You can now use the OTel service.name attribute to customize the name of SDv1 services."*, read 11/03/2026.

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Prerequisites

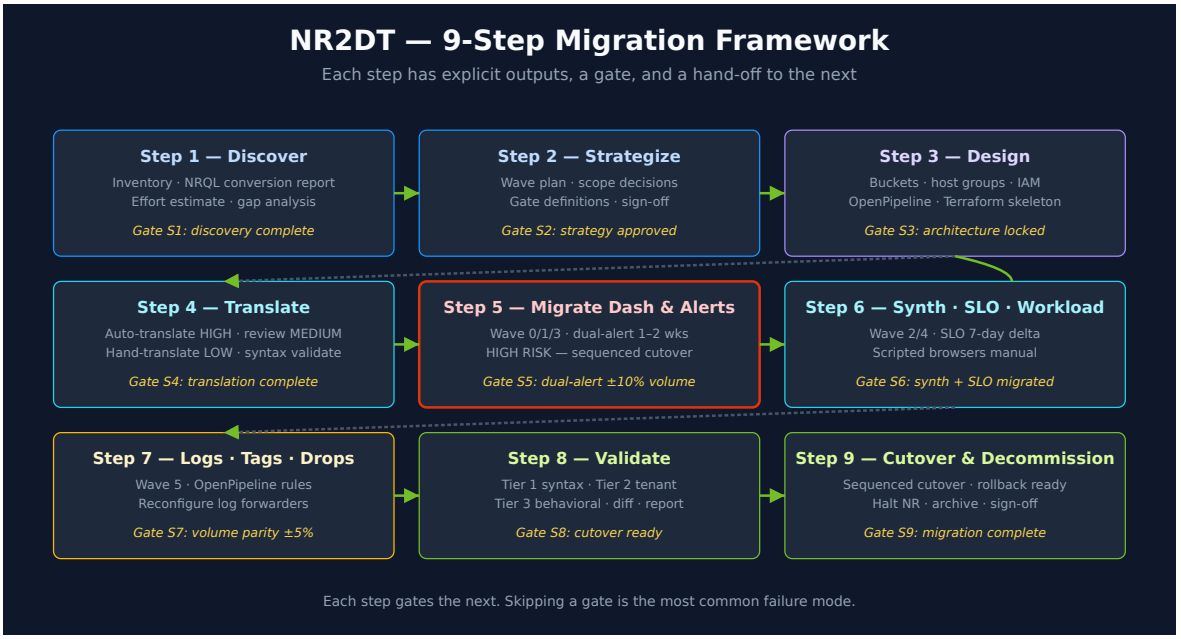
Requirement	Details
Audience	Migration lead + assigned engineer for this step
Format	Procedural step — use as a runbook; defer to NRLC for depth
NRLC deep dives	NRLC-01 (Platform Comparison) for orientation

1. What You'll Produce

By the end of this step you will have these artifacts checked into your migration repo:

Artifact	Format	Purpose
`inventory/exports/newrelic_export.json`	JSON	Export of the components named in `--components` (default: dashboards, alerts, synthetics, SLOs, workloads)
`nrql-conversion-report.csv`	CSV	Every NRQL query with HIGH/MEDIUM/LOW confidence + DQL output
`effort-estimate.md`	Markdown	Hours/days/weeks per component class
`gap-analysis.md`	Markdown	Entities flagged for manual work (APM conditions, scripted browsers)
`risk-register.md`	Markdown	Known risks with proposed mitigations

These artifacts gate Step 2 (Strategize). Don't proceed without them.



2. Prerequisites & Access

NR API key

A New Relic **User key** (`NRAK-…`) belonging to a user who has read access to the account's dashboards, alerts, NRQL data, synthetics, SLOs, workloads and logs. User keys carry no per-capability scopes — the key acts with its user's access: *"The user key allows you to make queries for any accounts you've been granted access to"*.

```
export NEW_RELIC_API_KEY="NRAK-..."
export NEW_RELIC_ACCOUNT_ID="&amp;lt;numeric&amp;gt;"
export NEW_RELIC_REGION="US"  # or EU
```

The tool reads exactly these names. `NR_API_KEY`, `NR_ACCOUNT_ID` and `NR_REGION` are not read, and every run then stops with `Configuration error`.

DT platform token

The tool validates **all four** variables at startup — the two `NEW_RELIC_*` values above and the two Dynatrace values below — on every `migrate` run, including the `--export-only` inventory in §3, and on `preflight` and `audit`. Set the Dynatrace values before the first run even though only §5 and Steps 5–8 actually connect to the tenant:

```
export DYNATRACE_API_TOKEN="dt0s16.XXXXX..."  # Platform Token
export DYNATRACE_ENVIRONMENT_URL="https://&amp;lt;env-id&amp;gt;.live.dynatrace.com"
```

`dt0s16` is the Platform Token prefix (*"Platform token enabling programmatic access to Dynatrace platform services"*); `dt0c01` is a classic access token.

Store these in `.env` rather than the shell for long-running work. Verify with:

```
python3 migrate.py preflight
```

`preflight` probes the tenant for Settings 2.0 / Document API / Automation API availability. Green here means Step 5+ will actually connect.

Tooling

Clone the migration framework:

```
git clone https://github.com/timstewart-dynatrace/NewRelic-to-Dynatrace-Migration-Utilities
cd NewRelic-to-Dynatrace-Migration-Utilities
pip install -r requirements.txt
cp .env.example .env  # populate NEW_RELIC_* and DYNATRACE_* vars
```

> **Sources:** New Relic API keys (New Relic docs), Tokens and authentication (DT docs), config/settings.py @ 78cbfce (tool repo, GitHub), read 10/05/2026.

3. Run the Inventory

Command:

```
python3 migrate.py migrate --export-only --output ./inventory
```

This runs the NerdGraph enumeration for the five default components — dashboards, alerts (exported as `alert_policies`), synthetics, SLOs and workloads — and writes one consolidated file, `inventory/exports/newrelic_export.json`. Add `--components dashboards,alerts,synthetics,slos,workloads,notification_channels` to include notification channels. At tool v3.0.0 (commit `78cbfce`) the export has no branch for drop rules, log-parsing rules or tag rules; count those in New Relic directly and record them by hand.

Check the output:

```
ls -lh inventory/
cat inventory/exports/newrelic_export.json | jq '{dashboards: (.dashboards|length), alert_policies: (.alert_policies|length), synthetic_monitors: (.synthetic_monitors|length), slos: (.slos|length), workloads: (.workloads|length)}'
```

Expected counts to capture (your numbers, for the wave plan in Step 2):

- Dashboards (and total widget count across all pages)
- Alert policies + NRQL conditions + APM conditions
- Synthetic monitors by type (Ping / Browser / API / Step)
- SLOs
- Workloads

- Notification channels by type
- Drop rules + log parsing rules + tag rules (not in the export — counted in New Relic by hand)

Extract NRQL for the translation pass

`compile` and `batch` (Step 4) need a flat list of NRQL queries. The export JSON has the NRQL embedded inside dashboards, alert conditions, and SLO indicators — use the `extract-nrql` subcommand to pull them into a file:

```
# Plain text (one NRQL per line — feeds `compile --file`):
python3 migrate.py extract-nrql \
    --input ./inventory \
    --output ./inventory/all-nrql.txt

# Or CSV with an `nrql` column (feeds `batch --file`):
python3 migrate.py extract-nrql \
    --input ./inventory \
    --output ./inventory/all-nrql.csv
```

The subcommand walks dashboards, alert conditions, and SLO indicators; dedupes; writes either format depending on the output extension (override with `--format txt|csv`). Capture the query count from the success line — that's the denominator for Step 4's confidence distribution.

▶ **Fallback for older tool versions** (without `extract-nrql`)

4. Translate the NRQL Surface

Every dashboard widget, alert condition, and SLO indicator contains a NRQL query that must compile to DQL. Run the compiler now to size the translation effort.

Command (uses the `inventory/all-nrql.txt` or `inventory/all-nrql.csv` produced by `extract-nrql` in §3):

```
# (a) compile — writes translated DQL, one per input query
python3 migrate.py compile --file inventory/all-nrql.txt --output nrql-translated.dql

# (b) batch — writes a CSV with nrql / dql / confidence / warnings columns.
#     Feed the .csv you created in §3 directly (or convert a .txt with awk):
python3 migrate.py batch --file inventory/all-nrql.csv --output nrql-conversion-report.csv
```

(If you extracted to `.txt` in §3 instead of `.csv`, convert with: `awk 'BEGIN{print "nrql"} {print "\"" $0 "\""}' inventory/all-nrql.txt > inventory/all-nrql.csv`)

Which artifact matters for the S1 gate? The CSV from `batch` — it has the confidence distribution you need to compare against the table below.

Output:

- `compile --output <file>` writes a plain DQL file (one translated query per input).
- `batch --output <file.csv>` writes a CSV with one row per query and exactly four columns: `nrql`, `dql`, `confidence` (HIGH/MEDIUM/LOW, or `FAILED` when the query did not compile), `warnings`. **This is the artifact to review for the Step 1 gate.**

Expected confidence distribution for a typical NR account:

Confidence	Typical share	Action in Step 4
HIGH	70–80%	Auto-translate; sample-validate during dual-run
MEDIUM	15–25%	100% review during dual-run
LOW	0–10%	Hand-translate or reformulate

If LOW is > 15%, surface this to stakeholders early — it's the dominant cost driver.

Deep-dive reference: NRLC-02 explains the compiler architecture, the 292 patterns, and how confidence is scored.

5. Generate the Effort Estimate

5a. Run the report generator (connects to DT, imports nothing)

```
python3 migrate.py migrate --dry-run --report --output ./discovery-run
```

> `--dry-run` **is what keeps this a preview.** Without it, `migrate` exports, transforms **and imports** the default components (dashboards, alerts, synthetics, SLOs, workloads) into the tenant during Discovery — `--report` runs after the import, it does not replace it.

`--output` is a **directory**, and `--input` is not read on this path. The report lands in `./discovery-run/reports/conversion-report.json` and `.html`: an NRQL→DQL conversion report (confidence summary plus per-query detail) built from the run's transform warnings. It is an input to `effort-estimate.md`, not the estimate itself.

> **Requires DT credentials** — `DYNATRACE_API_TOKEN` and `DYNATRACE_ENVIRONMENT_URL` from §2. Unset, the command stops with `Configuration error`; set but wrong, it stops with "Failed to connect to Dynatrace".
>
> **Sources:** migrate.py @ 78cbfce (tool repo, GitHub), read 10/05/2026.

5b. Offline fallback (no DT connection required)

If your Step 1 run happens before DT access is provisioned — typical for the first discovery sprint — derive the estimate offline from the inventory + conversion CSV:

```
python3 - &lt;&lt; 'PY' &gt; effort-estimate.md
import json, csv, pathlib

inv =
json.loads(pathlib.Path('inventory/exports/newrelic_export.json').read_text())
rows = list(csv.DictReader(open('nrql-conversion-report.csv')))

print('# NR→DT Migration — Discovery Artifact (Step 1)\n')
print('## Inventory counts\n')
for k in ('dashboards', 'alert_policies', 'synthetic_monitors', 'slos',
'workloads'):
    print(f'- **{k}**: {len(inv.get(k, []))}')
print('\n## Translation confidence distribution\n')
conf = {}
for r in rows: conf[r["confidence"]] = conf.get(r["confidence"], 0) + 1
total = sum(conf.values())
for k, v in sorted(conf.items()):
    pct = 100 * v / total if total else 0
    print(f'- **{k}**: {v} ({pct:.0f}%)')
print(f'\n## Totals\n\n- NRQL queries: {total}')
PY

cat effort-estimate.md
```

5b writes `effort-estimate.md` directly; with 5a, summarize the report into `effort-estimate.md`. Either satisfies the S1 exit criterion. Use 5a when DT access is ready; use 5b during early discovery.

Estimating heuristic per component class:

Class	Light (hrs)	Moderate (days)	Heavy (weeks)
Dashboards	< 20	20–80	80+
Alert conditions	< 50	50–200	200+
Synthetics (incl. scripted)	< 10	10–40	40+
SLOs	< 5	5–20	20+
Notification channels (with secrets)	< 5	5–20	20+

6. Step Exit Criteria

S1 — Discovery Complete

Pass criteria — all five must be true:

- [ ] `inventory/exports/newrelic_export.json` contains every component you exported, with no `✗ Failed to export` line in the run output (an empty array is acceptable when the NR account has none of that class); drop-, parsing- and tag-rule counts recorded by hand
- [ ] Per-component counts captured and documented
- [ ] `nrql-conversion-report.csv` generated; LOW count <= 15% of total
- [ ] `effort-estimate.md` reviewed by lead
- [ ] `gap-analysis.md` reviewed; manual-work items have estimated owners

If any check fails, do not proceed to Step 2. Fix the gap first.

Next step: NR2DT-02 — Strategize (wave planning, scope decisions, gate definitions for the rest of the migration).

This notebook was AI-generated from community-submitted and publicly available sources, including the open-source NewRelic-to-Dynatrace-Migration-Utilities, nrql-engine, and nrql-translator projects. This notebook series is not officially supported by Dynatrace or New Relic. Always verify information against the official Dynatrace documentation and New Relic documentation.